

Classwork 1

Powers, factorising and algebraic fractions

Grades: [A] [M] [E]

Show all working. Leave answers with positive indices and fractions in simplest form.

Part A · Powers, roots and factorising

A1 Simplify $\frac{(5x^2)^2}{10x^{-3}}$, leaving your answer with positive indices. [A]

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A2 Simplify $\sqrt[3]{125p^9}$. [A]

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A3 Factorise completely $9x^2y - 6xy^2$. [A]

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A4 Factorise $2x^2 - x - 15$. [A]

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A5 Simplify $\frac{x^2 - 16}{x^2 + 2x - 8}$. [A]

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Part B · Combine, solve, change the subject

B1 Write $\frac{3}{x+2} - \frac{x-4}{x^2+2x}$ as a single fraction in its simplest form. [M]

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B2 Solve $\frac{3}{x} + \frac{1}{x-2} = 2$. [M]

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B3 Make t the subject of $m = \frac{3t - 1}{t + 2}$.

[M]

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Part C · Show that

C x is a whole number. Show that $(x + 1)^3 - (x - 1)^3$ is always **2 more than a multiple of 6**.

[E]

Words you may use

expand · collect like terms · multiple of 6 · whole number · therefore

Sentence starters (optional): *Expanding the first bracket gives ... Subtracting, ... Since x^2 is a whole number,*
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Extension

Find every whole number x for which $(x + 1)^3 - (x - 1)^3 = 56$.

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